Kai Barker

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EDUCATION

University of California, Santa Barbara Santa Barbara, CA
Statistics and Data Science B.S. Graduating June 2026
Major GPA: 3.90/4.0
Activities: Consult Your Community, Statistics & Data Science Club, Data Science Collaborative, Ski & Snow Club
Coursework: Linear Algebra, Optimization, Machine Learning, Stochastic Processes, Time Series, Adv. Statistical Models, Big Data

SKILLS, CERTIFICATES & INTERESTS

Languages & Tools: Python (Pandas, Numpy, etc.), R, SQL, Java, C++, HTML, CSS, Tableau, Excel, Git, API Integration, Spark, LaTeX
Skills: Statistical Modeling, Time Series Forecasting, Feature Engineering, Machine Learning (Scikit, PyTorch), Pipeline Development
Certificates: JP Morgan Chase Software Engineering, JP Morgan Chase Quantitative Analysis, HackerRank SQL
Interests: Basketball, Hiking, Backpacking, Rock Climbing, Cooking, Running, Pickleball

EXPERIENCE

Engagement Manager Santa Barbara, CA
Consult Your Community February 2026 – Present

- Built a statewide financial model across all 115 CA community colleges using SCFF apportionment data, identifying that a 12% reduction in gateway math DFW rates generates $26M in recoverable state funding
- Designed a Tableau dashboard visualizing DFW rates, math success, and recoverable funding across CA community college market
- Led a 16-week EdTech market entry engagement for an AI tutoring startup, directing a 7-person team across 3 research workstreams to deliver a market entry strategy and value proposition targeting the client's first paid B2B pilot

Business Analyst Santa Barbara, CA
Consult Your Community October 2025 – Present

- Produced acquisition readiness assessment across operations, financials, and marketing for a small business client; identified 3+ areas to reduce owner dependency ahead of sale.
- Diagnosed over-reliance on owner relationships and gaps in client follow-up; recommended CRM and communication improvements projected to increase annual revenue by $40,000 and improve business attractiveness to buyers.
- Built a Tableau dashboard with 10+ visualizations to highlight operational gaps, driving recommendations presented to stakeholders.

Quantitative Researcher Santa Barbara, CA
Department of Statistics and Applied Probability September 2024 – Present

- Analyzed 80,000+ congressional trade disclosures to identify statistically significant abnormal returns, validating a weighted alpha signal for portfolio construction.
- Applied Random Matrix Theory (Marchenko-Pastur) and Ledoit-Wolf shrinkage to clean a 560-asset covariance matrix, isolating 30 signal eigenvalues explaining 60.1% of variance.
- Optimized mean-variance portfolios in Python using cvxpy, achieving a 13.81% annual backtest return and +1.07% mean alpha (excluding 2022) vs. SPY over an 8-year walk-forward period.

Research Assistant Santa Barbara, CA
Foundations of Financial Technology Lab September 2024 – Present

- Analyzed 20M+ Ethereum blocks and 100M+ wallets to quantify wealth distribution, calculated a Gini coefficient of 0.96.
- Produced visualizations tracking inequality evolution over 5 years, revealing the top 1% of wallets control 95%+ of Ether supply.
- Filtered institutional and Sybil addresses to isolate organic wallet behavior and improve the accuracy of distribution metrics.
- Acknowledged in a published paper on blockchain wealth inequality with researchers from Yale, Cornell, and UCSB.

PROJECTS

NeurIPS Ariel Data Challenge
Competition - ESA Ariel Mission

- Built a signal processing pipeline (ADC conversion, sigma-clipped pixel masking, dark and flat field correction) to calibrate 200+ GB of raw telescope data across 14,000 files and 1,100+ exoplanets.
- Engineered a weighted ensemble of Gaussian Process Regression (composite RBF + Matern kernel), XGBoost, and Ridge to predict atmospheric spectra across 283 wavelength channels.
- Calibrated prediction uncertainty via GLL-optimal grid search over scale parameters, achieving a final score of 0.311.

MCMC & Simulated Annealing - Optimization Research
Academic Research

- Implemented Metropolis-Hastings MCMC in Python for message decryption, constructing the target likelihood from letter-to-letter transition probabilities trained on War and Peace text.
- Extended the algorithm with a simulated annealing modification (exponential cooling schedule, alpha=0.995) over 50,000 iterations, reducing average incorrect letters from 22.3 to 21.6 vs. baseline.